INTRODUCTION

Module leader: **Alessandro Di Stefano**, Senior Lecturer in Computer Science at Teesside University, within the School of Computing, Engineering and Digital technologies (SCEDT), Department of Computing & Games. For further information about my research profile, please visit the following website: https://research.tees.ac.uk/en/persons/alessandro-di-stefano

Artificial Intelligence (AI) Foundations (CIS4049-N) module provides students with the foundational knowledge to study a wide range of AI applications and solutions. The module introduces knowledge representation, reasoning, problem solving and algorithms, planning and several AI applications.

Students will be assessed by a written report that investigates the implementation of AI techniques to a case study of their choice and critically evaluate the selection, implementation, and experimental validation of the results.

A detailed **Teaching Schedule** is provided on Blackboard.

Please see the Schedule link under Module Information and Resources.

MODULE INFORMATION

https://apps.tees.ac.uk/ModuleCatalogue/View.aspx?SITSCode=CIS4049-N&ACYear=2021/2

THE MODULE IS DESIGNED TO:

This module aims to:

1. Provide an introduction to the foundations of Artificial Intelligence (AI)
2. Introduce AI concepts, applications, and theory
3. Explore the key factors that have made the AI successful for various applications
4. Give an awareness of the current state and direction of AI research

THE LEARNING OUTCOMES

At the end of the module, you will be able to:

PERSONAL AND TRANSFERABLE SKILLS:

1. Effectively communicate and evaluate complex information related to AI theory.
2. Use personal reflection to analyse self and own actions in the course of working as a group or individual in a real-world AI case study.
3. Reflect upon, take ownership of, and critically appraise the outcome of an implemented solution against a given brief for a simulated or real-world problem using appropriate AI technologies.

RESEARCH, KNOWLEDGE AND COGNITIVE SKILLS:

4. Critically analyse a solution to a given real world case study.
5. Utilise effectively AI software and techniques in problem solving.
6. Critically appraise of recent scientific literature in AI in the context of a given scenario.

PROFESSIONAL SKILLS:

7. Critically evaluate the commercial risks and opportunities related to solving an AI problem.

8. Autonomously evaluate improvements to performance drawing on innovative or best practice in applied AI related skills.

FORMAT OF MODULE

Lecture: 1 hour; IT Labs: 2 Hours

Subject areas will be discussed in lectures and further underpinned by practical sessions which will provide the opportunity to apply the theory to real world problems. Medium-scale pieces of practical work, based on a problem-based learning approach, will be set to accompany lecture sessions. Students will be expected to complete much of this practical work in their own time, but lab sessions will also be used to provide support to students as they develop practical solutions and undertake their in-course assessed work.

Formative feedback on student progress will be offered each week in the lab-based practical sessions. The results of the assignment are presented to peers and academics towards the end of the module.

The students will be expected to develop their reflective development skills to encourage a deep approach to their learning.

The total number of learning hours for the module is **200**, the lectures and tutorials together only account for about **36** hours of this. The lectures and tutorials are designed to sign post areas of importance. You must therefore do extra work, including reading books recommended (please see bibliography section) and build upon the skills developed in the lab sessions.

E-LEARNING @TEES (BLACKBOARD VIRTUAL LEARNING ENVIRONMENT)

The module will be supported by Blackboard. When you log into Blackboard, the module should be listed. If it is not, please contact the module leader.

Blackboard can be found at the following link: https://bb.tees.ac.uk/ultra/

MODULE ASSESSMENT AND FEEDBACK

The assessment will be in the form of an in-course assignment (ICA) that requires students to design and implement an AI solution selecting and using appropriate AI techniques.

Assessment will be a single individual in-course assessment (ICA), which requires the students to produce a written report and a walkthrough video.

In the written report, students will critically evaluate and reflect on the application of AI techniques to a real-world case study of their choice and justify the utilisation of these techniques (equivalent to 4,000 words).

Moreover, the student is also required to produce a voice over brief walk-through video (2-3 minutes), showing, and demonstrating what has been done in the implementation. It is expected that the student will introduce his work and discuss what has been achieved. It is also recommended that the student highlights the issues and limitations encountered during implementation. This in-course assessment will meet all the learning outcomes: PTS1, PTS2, PTS3, RKC1, RKC2, RKC3, PS1, PS2.

The School operates a standard procedure for providing (at least) a minimum level of feedback to students in line with the University's Assessment and Feedback Policy.

For more details about the assessment, read the ICA documentation on Blackboard, in the Assessment link.

Students will be given feedback on their progress via review of their work in lab sessions.

MARKING CRITERIA

You will be provided with specific assessment marking criteria within the ICA. This will indicate how to achieve the stated learning outcomes for the module. Please see the Assessment link within the main menu on Blackboard.

REFERENCING

Please note that the University uses a standard form for referencing based on the Harvard approach, full details of how to reference correctly using this standard can be found in:

Pears, R. and Shields, G. (2016). *Cite them right: the essential referencing guide.* 10th edn. Basingstoke: Palgrave Macmillan. Copies of this text are available in the Library and online at: on-campus at http://www.citethemrightonline.com/

or for off-campus access use http://ezproxy.tees.ac.uk/login?url=http://www.citethemrightonline.com/

There is also additional support available in the library: Learning Hub guidance on referencing: http://libguides.tees.ac.uk/referencing

READING LIST & RESOURCES

Please see Reading List Online within the module on Blackboard.

BOOKS – AVAILABLE ONLINE

Russell, S. and Norvig, P., 2020 - 4th edition. *Artificial Intelligence: A Modern Approach*. Pearson Education (US) ISBN-13: 9780134610993.

Eckroth, J., 2018. *Python Artificial Intelligence Projects for Beginners.* Packt Publishing Limited, ISBN-13: 9781789538243

ELECTRONIC

Python for Artificial Intelligence: https://wiki.python.org/moin/PythonForArtificialIntelligence

Ponteves, Hadelin de, 2019. *AI Crash Course*. Packt Publishing Limited. ISBN-13: 9781838645557

Bengio, Y., Goodfellow, I. and Courville A., 2016. *Deep Learning*. MIT Press Ltd Cambridge, Massachusetts: http://www.deeplearningbook.org